Exercise 4

1.1

```
Select  EmployeesID,
        FirstName,
        Last_Name,
        Department,
        Salary,
        Project ID,
        ProjectName,
        Budget,
        Status
From  Employees AS A
INNER Join  Projects AS B.
ON  A.EmployeeID = B.EmployeeID;
```

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
1	John	Doe	IT	70 000	103	Cyber Security Audit	75 000	Pending
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
5	Emma	Wilson	Sales	65 000	105	Market Expansion	65 000	Completed
6	Michael	Clark	Finance	80 000	106	Risk Management	80 000	Pending

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1.2) Select Employee,
First Name,
Last Name,
Department,
Salary,
Project ID,
Project Name,
Budget,
Status

From Employees AS A
Left Join Projects AS B.
On A. Employees ID = B. Employees ID;

Employee ID	First Name	Last Name	Department	Salary	Project ID	Budget	Project Name	Status
1	John	Doe	IT	70 000	101	100 000	AI Development	Comp
1	John	Doe	IT	70 000	103	75 000	Cyber Security Audit	Pend
2	Alice	Smith	HR	60 000	102	50 000	Employee Training	Ongoi
3	Bob	Johnson	Finance	75 000	104	90 000	Financial Analysis	Ongoi
4	David	Brown	IT	72 000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65 000	105	65 000	Market Expansion	Compl
6	Michael	Clark	Finance	80 000	106	80 000	Risk Management	Pendi

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1.3 Select ProjectID,
ProjectName,
Budget,
Status,
EmployeeID,
FirstName,
LastName,
Department,
Salary
From Employees As A
Right Join Projects As B
ON A.EmployeeID = B.EmployeeID;

Employee ID	FirstName	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development		Completed
2	Alice	Smith	HR	60 000	102	Employee Training		Ongoing
1	John	Doe	IT	70 000	103	Cybersecurity Audit		Pending
3	Bob	Johnson	Finance	75 000	104	Financial Analysis		Ongoing
5	Emma	Wilson	Sales	65 000	105	Market Expansion		Completed
6	Michael	Clark	Finance	80 000	106	Risk Management		Pending

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1.4 Select EmployeeID,
FirstName,
Last Name,
Department,
Salary,
ProjectID,
ProjectName,
Budget,
Status
From Employees AS A
Full outer Join Projects AS B.
ON A.EmployeeID = B. EmployeeID;

Employee ID	FirstName	Last Name	Department	Salary	Project ID	Project Name	Budget	St
1	John	Doe	IT	70000	101	AI Development	100000	Comp
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pend
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongo
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongo
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market expansion	60000	Comp
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pend

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3.7) Select EmployeeID,
FirstName,
LastName,
Department,
Sellary
From Employees
Where Salary > 70 000;

Employee ID	First Name	Last Name	Department	Sellary
3	Bob	Johnson	Finance	75 000
4	David	Braun	IT	72 000
6	Michael	Clark	Finance	80 000

3.8 Select EmployeeID,
First Name,
Last Name,
Department,
Sellary
From Employees
Where Department IN ('IT', 'Finance');

Employee ID	First Name	Last Name	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75 000
4	David	Braun	IT	72 000
6	Michael	Clark	Finance	80 000

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3.9) Select Project ID,
Project Name,
Budget,
Status
From Projects
Where Status != 'Completed';

Project ID	Project Name	Budget	Status
102	Employee Training	50 000	Ongoing
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

3.10 Select Project_ID,
Project Name,
Budget,
Status
From Projects
Where Budget > 70 000 and Status != Completed

Project ID	Project Name	Budget	Status
103	Cyber Security Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

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3.11 Select EmployeeID,
First Name,
Last Name,
Department,
Salary,
City
From Employees
Where City IN ('New York', 'Toronto')
Order by Salary DESC;

EmployeeID	First Name	Last Name	Department	Salary	City
6	Michael	Clark	Finance	80 000	New York
3	Bob	Johnson	Finance	75 000	Toronto
1	John	Doe	IT	70 000	New York

3.12 Select Employee ID,
First Name,
Last Name,
Department,
Salary.
From Employees
Order by Salary DESC
Limit 3;

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000

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4.13. Select Department,
SUM(Salary) AS TotalSalary
From Employees
Group by Department
Order by TotalSalary DESC;

Department	TotalSalary
Finance	155 000
IT	142 000
Sales	65 000
HR	60 000

4.14 Select City,
AVG(Salary) AS Average Salary
From ~~Employees~~ Projects Employees
Having ~~AVG(Salary)~~ Average Salary > 65 000;

City	Average Salary
New York	75 000
Toronto	75 000

4.15 Select Department,
~~Count(Department)~~
Count(Department) AS EmployeeCount
From Employees
Having EmployeeCount > 1

~~Dpa~~ Department	EmployeeCount
IT	2
Finance	2

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4.16 Select ~~Project~~ Status,
Count (Status) As ProjectCount

From Projects
Where ProjectCount >= 2

Status	ProjectCount
Ongoing	2
Pending	2

4.17 Select EplayeeID,
FirstName,
LastName,
Sum(Budget) AS TotalProjectBudget
From Employees AS A
Join Projects AS B
ON A.EmployeeID = B.EmployeeID
Having TotalProjectBudget > 150000;
~~[illegible]~~

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175 000

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